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DomainDiff: Unified Two-Stage Optimization for Text-Video Retrieval

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Abstract

The primary challenge in text-video retrieval lies in achieving cross-modal semantic alignment, particularly the discrepancy between the conciseness of textual descriptions, which often fail to fully encapsulate the breadth of video content, and the redundancy in video data, which introduces noise and masks important semantic features. Current methods align text and video by mapping them into a shared feature space. Despite notable advancements, the inherent differences in modality-specific representations create a bottleneck for fixed-point embedding techniques, making models highly sensitive to dataset distribution and hindering their generalization ability. In this paper, we present DomainDiff, a framework that enhances the embedding space through a two-stage process. In the first stage, stochastic domain modeling, we semantically expand text embeddings to explore potential regions aligned with video content. Simultaneously, we filter video segments to reduce redundancy and highlight key frames. In the second stage, the dynamic agent attention diffusion network, we leverage the generative properties of diffusion models to optimize the embedding space by viewing it from a joint probability distribution perspective. An agent attention mechanism dynamically integrates text and video features, ensuring accurate cross-modal alignment. Experimental results demonstrate that DomainDiff significantly improves retrieval performance across five benchmark datasets, with R@1 improvements ranging from 3% to 7.4%. Moreover, DomainDiff outperforms existing methods in handling long videos and complex textual descriptions, showcasing superior semantic robustness and generalization across varying distributions.

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• Information systems → Multimedia and multimodal retrieval.

Keywords

Text-Video Retrieval, Multimodal Alignment, Representation Learning, Agent Attention Mechanism, Diffusion Model

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1 Introduction

Text-Video Retrieval aims to retrieve the most semantically relevant video segments (or textual descriptions) from a candidate set given a text query (or a video segment) [4, 6, 10, 48]. Due to the inherent differences between video and text modalities, achieving precise cross-modal retrieval alignment remains a significant challenge. Existing research primarily focuses on two directions. On the one hand, researchers attempt to apply powerful vision-language models to the field of video-text retrieval [18, 49]; on the other hand, specific modules [9, 56, 63] are designed for fine-grained semantic extraction of videos or texts, such as temporal modeling [26, 55]. Furthermore, introducing new modalities (for example, audio [1, 24, 38, 46]) and performing feature fusion before or during the alignment stage has become a new research trend [8, 18, 26].

Mainstream text-video retrieval methods mainly focus on extracting more precise textual and video features and projecting them into a shared feature space for alignment [49, 53, 57, 60]. Although some methods have achieved significant performance improvements, the strategy of embedding text and video into a single fixed feature point has limitations [44, 60]. This fixed-point embedding approach is sensitive to variations in the dataset, resulting in limited model robustness and reduced generalization ability. Through observations of mainstream datasets, we find that most textual descriptions are overly concise [37, 53, 59], often presented as labels as shown in Figure 1 (a). Although videos may contain redundant information [15, 18, 62], especially long videos often encompass a wealth

of valuable information. In the alignment approach of projecting modalities into a shared feature space [44, 49, 60], textual embeddings often fail to comprehensively cover the core semantics of videos, while video embeddings struggle to accurately represent their rich information.

From a probabilistic perspective, the alignment approach based on a shared feature space essentially models the conditional probability distribution $p(\text{candidates}|\text{query})$. This method cannot effectively model the underlying data distribution $p(\text{query})$ [5, 28, 36], resulting in a feature space that lacks intrinsic data characteristics and thus struggles to achieve good generalization on unseen data.

To address these issues, we propose a novel Domain Diffusion (DomainDiff) cross-modal retrieval framework that dynamically samples the positions of text and video embeddings while continuously refining the sampling domain via a diffusion network. By repeatedly sampling within the domain, our approach facilitates semantic exploration and progressively enriches the original text’s semantic content, simultaneously eliminating redundant video frames and highlighting key semantic ones. Consequently, the text and video embedding points are drawn closer together in the feature space, as illustrated in Figure 1 (b).

Our framework includes two stages: Stochastic Domain Modeling and Dynamic Agent Attention Diffusion Network. Each stage targets distinct objectives to jointly refine the text-video embedding space. In the first stage, we perform stochastic domain sampling on both textual and video embeddings to identify superior embedding points. To address the challenge of short texts failing to capture the rich semantics of videos [37, 53], potential matching regions are explored to supplement the original textual embeddings. At the same time, to mitigate the impact of redundant video frames on the core semantic representation [15, 62], frames or segments that closely align with the text are selectively filtered through stochastic sampling. In the second stage, the stochastic text-video embeddings obtained previously are fed into a dynamic agent diffusion network. This network dynamically adjusts the embedding positions through an agent attention mechanism, achieving a finer-grained cross-modal alignment. Moreover, by leveraging the generative capabilities of diffusion models, the embedding space is further optimized from the perspective of a joint probability distribution.

We observe that the text-based stochastic domain sampling process can effectively uncover high-matching embedding locations for videos, enriching the original semantic information. This process generally stays within the video’s rich semantic range, effectively identifying frames or semantics most relevant to the text, enabling a core semantic replacement of the video’s original representation. Furthermore, the text-video domain embeddings obtained in the first stage help to better activate the potential of the diffusion model.

The main contributions of this paper are as follows:

1. We propose the DomainDiff cross-modal retrieval framework, which consists of two stages: Stochastic Domain Modeling and Dynamic Agent Attention Diffusion Network. These stages work together to optimize the text-video embedding space.

2. By performing stochastic domain sampling on textual embeddings, potential matching regions are explored to supplement the semantics of short texts. By performing stochastic domain sampling on video embeddings, high-matching frames are filtered to reduce redundancy interference and enhance core semantic representation.

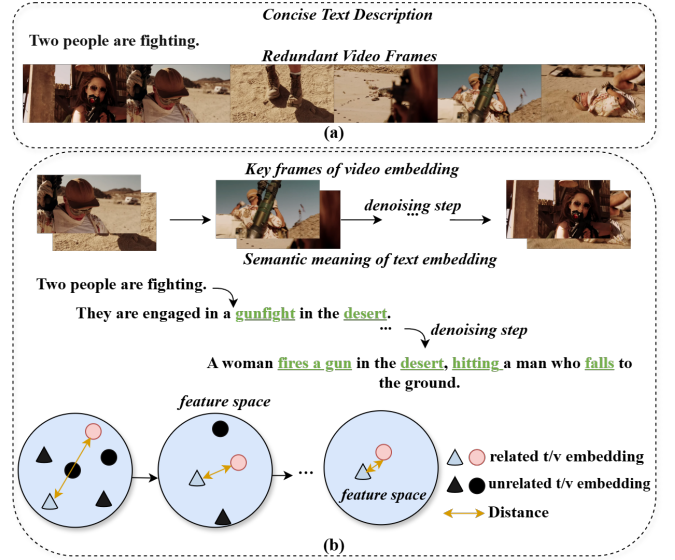


Figure 1: (a): One major issue with datasets is the conciseness of text descriptions and the redundancy in video content. (b): With DomainDiff two-stage optimization, text embeddings capture richer semantic details, while video embeddings highlight the key representative frames of an entire video. This collaborative design reduces the distance between text and video in the feature space.

3. Based on the modal embedding optimization in the first stage, the joint probability distribution is modeled using diffusion models, significantly enhancing model generalization capabilities, reducing dataset dependency, and improving adaptability to unseen data.

4. Experimental results demonstrate that the proposed framework significantly improves performance on five benchmark datasets (MSRVTT [59], LSMDC [50], DiDeMo [3], ActivityNet [31], and VATEX [54]), achieving state-of-the-art results (R@1 improvements of 3% ~ 7.4% over baselines).

2 Related Work

Text-Video Retrieval Current mainstream datasets (MSRVTT [59], DiDeMo [3], LSMDC [50]) suffer from two inherent limitations: text descriptions are overly concise [37, 53] and fail to comprehensively cover video content, and videos contain redundant information [15, 18, 62] that complicates core semantic feature extraction, leading to suboptimal text-video matching accuracy. To address this challenge, on the one hand, researchers have attempted to apply powerful vision-language large models in the field of text-video retrieval. For example, methods such as UMT [32], VALOR [39], and vid-TLDR [11] employ large models for unified multimodal tasks to enhance performance; on the other hand, some approaches design dedicated modules to extract fine-grained semantic features from videos or texts. For instance, UCoFiA [57] and HBI [26] further improve alignment by combining coarse- and fine-grained features, while VAST [8] and VALOR [39] enhance alignment by incorporating additional modalities. Despite their progress, these approaches essentially optimize *fixed embedding representations* (e.g., through CLIP’s frozen parameters or limited fine-tuning), struggling to

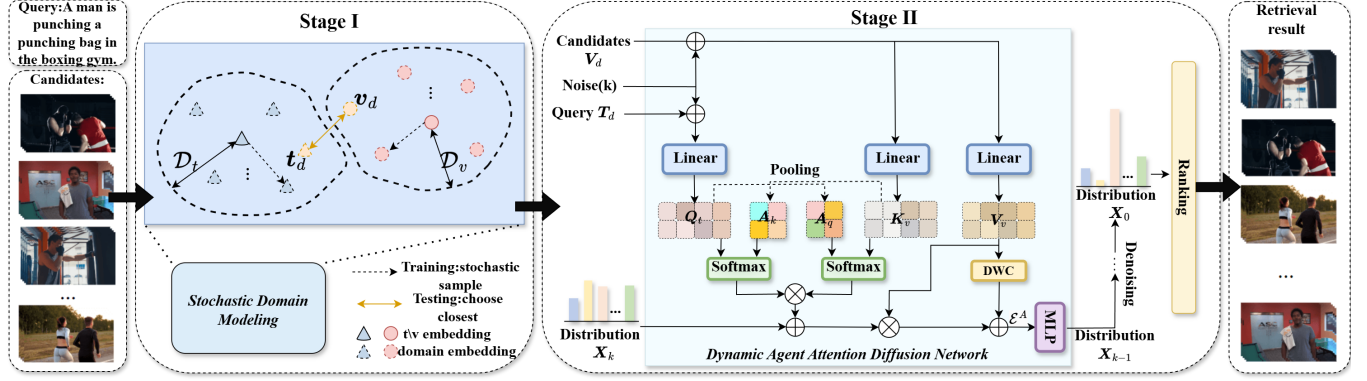


Figure 2: Illustration of the DomainDiff Framework. In Stage I, initial text–video embeddings are stochastically sampled from their respective domains ($\mathcal{D}_t$ and $\mathcal{D}_v$, Eq. (5)) to generate domain-specific embeddings (t_d and v_d , Eq. (3)), which are concatenated into T_d and V_d . In Stage II, these embeddings, along with Noise(k) and distribution X_k , are fed into the Dynamic Agent Attention Diffusion Network (DAADN). DAADN applies dual-stream attention between projected features and learned agent tokens to model a noisy data distribution ($\mathcal{E}^A$, Eq. (10)), which is refined via a denoising MLP to recover the aligned distribution X_0 .

dynamically adapt to complex semantic distribution [37, 53]. Recent efforts attempt to integrate denoising diffusion techniques for video representation optimization (e.g., DiffusionRet [28], MomentDiff [33]), yet their potential for dynamic embedding space modeling remains under explored.

Diffusion Models for Representation Learning From a probabilistic perspective, while contrastive learning models *conditional probability* i.e. $p(\text{candidates}|\text{query})$ through negative sample distance maximization, diffusion-based generative approaches directly model *joint probability distribution* i.e. $p(\text{candidates}, \text{query})$ via reverse denoising processes. This enables diffusion models to comprehensively capture intrinsic data characteristics i.e. $p(\text{query})$ and achieve more accurate latent space projection [5, 36], thereby exhibiting stronger cross-domain generalization than contrastive learning models. In visual self-supervised learning, DiffMAE [58] replaces MAE’s [22] L2 loss with a diffusion decoder, and DARL [35] trains autoregressive models using diffusion-driven patch decoders. For multi-modal tasks, DiffusionVMR [61] jointly optimizes video moment retrieval and boundary detection through denoising generation, while Luo et al. [43] edit source videos via text-guided diffusion to enhance unseen domain adaptability. Nevertheless, existing methods primarily treat diffusion models as auxiliary training modules, failing to fully exploit their dynamic embedding space optimization capabilities. To bridge this gap, we propose a two-stage hybrid training framework that deeply integrates diffusion-based joint distribution modeling into contrastive learning, enabling both global alignment and local semantic refinement for robust text-video retrieval.

3 Method

3.1 Problem Definition

Given a corpus $C = \{(t, v)\}_{i=1}^K$ containing K text-video pairs, where each text t consists of N words w and each video v contains M frames f , the goal of text-video retrieval is to retrieve the most semantically relevant video v from a video set V ($v \in V$) given a query text t . To achieve this, we aim to learn a text encoder ψ_t and a video encoder ψ_v , which map text and video into a shared

embedding space, thereby enabling cross-modal feature alignment. The overview of our DomainDiff framework is shown in Figure 2.

3.2 Stage I Stochastic Domain Modeling

We employ pre-trained text encoder ψ_t and video encoder ψ_v to extract fine-grained feature representations, respectively. Given a text t and a video v , their fine-grained embeddings can be represented as:

$$\mathbf{w}_i = \psi_t(t), \forall i \in [1, N]; \quad \mathbf{f}_j = \psi_v(v), \forall j \in [1, M], \quad (1)$$

where $\mathbf{w}_i, \mathbf{f}_j \in \mathbb{R}^d$ denote word-level and frame-level embeddings. The dimension of the feature space is d . N and M represent the number of words and video frames. By extracting the class token ([cls]) as global features,

$$\mathbf{c}_t = [\text{cls}]_w; \quad \mathbf{c}_v = [\text{cls}]_f \quad (2)$$

we obtain the global embeddings of text and video $\mathbf{c}_t, \mathbf{c}_v \in \mathbb{R}^d$.

To address the issue of overly concise textual descriptions and redundant video data affecting cross-modal alignment, we propose a stochastic sampling-based representation strategy in the latent domain, which projects text and video into stochastic representations during the embedding phase to reduce the distance between positive samples. Specifically, we employ the reparameterization trick [30]. For given text-video global feature vector $\mathbf{c}_t, \mathbf{c}_v \in \mathbb{R}^d$, where the feature space has a dimension of d , we introduce latent domains $\mathcal{D}_t$ and $\mathcal{D}_v$. We perform stochastic sampling within these domains to obtain stochastic domain embeddings $\mathbf{t}_d$ and $\mathbf{v}_d$ representing stochastic text and video embeddings respectively:

$$\begin{aligned} \mathbf{t}_d &= \mathbf{c}_t + \mathcal{D}_t \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1), \\ \mathbf{v}_d &= \mathbf{c}_v + \mathcal{D}_v \cdot \epsilon, \quad \epsilon \sim \mathcal{N}(0, 1), \end{aligned} \quad (3)$$

where $\epsilon \sim \mathcal{N}(0, 1)$ is used for stochastic sampling. $\mathbf{c}_t$ and $\mathbf{c}_v$ Eq. (2) are the global feature embeddings of the raw text and video.

Subsequently, the stochastic domain embeddings $\mathbf{t}_d$ and $\mathbf{v}_d$ are used to compute the cross-entropy loss Eq. (6). However, determining the stochastic domain range presents certain challenges. On the one hand, if the stochastic domain range $\mathcal{D}$ is too small, it may

lead to insufficient modal representation capability. On the other hand, if $\mathcal{D}$ is too large, it may introduce inappropriate embedding points, thereby degrading retrieval performance. To address this, we propose a Domain Similarity Awareness Module (DSAM). First, we compute the cosine similarity between text and video frames:

$$\begin{aligned} S_j^t &= s(\mathbf{c}_t, \mathbf{f}_j), \forall j \in [1, M], \\ S_i^v &= \frac{1}{M} \sum_{j=1}^M s_{ij}(\mathbf{w}_i, \mathbf{f}_j), \forall j \in [1, M], \forall i \in [1, N] \end{aligned} \quad (4)$$

where S_j^t denotes the similarity between the sentence(text global representation $\mathbf{c}_t$) and the j -th video frame, and S_i^v represents the average similarity between the i -th word and each video frame. Notably, the video global representation $\mathbf{c}_v$ is not used here as computing similarity based on video frames is more appropriate for word-level matching.

Based on these similarities, we compute the stochastic domain of text and video using learnable Linear Layer Θ , denoted $\mathcal{D}_t, \mathcal{D}_v$:

$$\begin{aligned} \mathcal{D}_t &= \exp(\mathbf{S}_t \Theta_t), \mathbf{S}_t = [S_1^t : S_M^t], \\ \mathcal{D}_v &= \exp(\mathbf{S}_v \Theta_v), \mathbf{S}_v = [S_1^v : S_N^v], \end{aligned} \quad (5)$$

where $\mathbf{S}_t$ and $\mathbf{S}_v$ are the domain similarity tensors for text and video, respectively. Additionally, we explore the use of a learnable scalar θ to compute the domain range: $\mathcal{D} = \exp\left(\frac{\theta}{M} \sum_{i=1}^M S_i\right)$, where θ is broadcast to d -dimensions to adapt to high-dimensional space. For more details, refer to Table 3. Specifically, as long as a point falls within the representation domain, it can be considered a valid representation of the modality and used for similarity computation.

During training, we incorporated $\mathbf{t}_d$ and $\mathbf{v}_d$ into the $\mathcal{L}_{ce}$ [49], achieving more robust modality alignment through a dynamic sampling mechanism. The corresponding cross-entropy loss function is defined as,

$$\mathcal{L}_{ce} = \frac{1}{2} \mathbb{E}_{(t,v) \in C} [\log p(v|t; \psi_t, \psi_v) + \log p(t|v; \psi_t, \psi_v)], \quad (6)$$

which primarily emphasizes conditional relationships between modalities. The conditional probability $p(v|t)$ as well as $p(t|v)$ is defined by the following formula:

$$p(v|t; \psi_t, \psi_v) = \frac{\exp(\psi_t(t)^\top \psi_v(v)/\tau)}{\sum_{v' \in V} \exp(\psi_t(t)^\top \psi_v(v')/\tau)}, \quad (7)$$

where the temperature hyperparameter τ regulates the similarity in the embedding space, and a normalized distribution is constructed over the candidate video set V . During evaluation, we obtain the optimal embedding points through multiple sampling iterations, ensuring the stability and accuracy of the representation.

Unlike traditional methods that directly use fixed-point embeddings in cross-entropy loss Eq. (6) to explicitly adjust the distance between text-video embedding points, our proposed dynamic stochastic embedding strategy based on domain range sampling enhances cross-modal representation capabilities by implicitly narrowing the distance between text-video embedding points. To overcome the limitation of concise texts in capturing the rich semantics of videos, we employ stochastic sampling to explore potential matching domain, thereby enhancing the semantic content of the original text embeddings. Likewise, to mitigate the interference of redundant

video information with core semantic representation, we use stochastic sampling to filter out frames or segments that exhibit a high degree of relevance to the text.

However, it is important to note that stochastic domain learning methods optimize model parameters by maximizing conditional probabilities, specifically $p(v|t; \psi_t, \psi_v) + p(t|v; \psi_t, \psi_v)$. A fundamental limitation of this approach is its inability to model the inherent distributions of text and video data, $p(t)$ and $p(v)$, particularly their underlying statistical characteristics and joint dependencies. This shortcoming is evident in Eq. (7).

This design paradigm inherently disrupts the intrinsic structure of the original data distributions, preventing the model from effectively capturing fundamental data patterns. Consequently, its cross-domain generalization ability is significantly hindered, limiting adaptability to unseen data.

3.3 Stage II Dynamic Agent Attention Diffusion Network

To further optimize text-video representation and address the issue in Stage I, where the inherent data distributions $p(t)$ and $p(v)$ are overlooked, we model the joint probability distribution $p(v, t)$ from a generative perspective. Specifically, we propose a Dynamic Agent Attention Diffusion Network (DAADN) to capture the statistical properties and latent dependencies of cross-modal data. The objective function is defined as:

$$\prod_{(t,v) \in C} p(v, t) \approx \phi(v, t, \mathcal{N}(0, \mathbf{I})), \quad (8)$$

where C represents the text-video pair corpus, and ϕ denotes the parameters of the diffusion network. Unlike generative tasks prioritizing diversity and fidelity [2, 7, 47], cross-modal retrieval demands precise modality alignment [45, 49]. Our framework achieves this through a novel fusion of diffusion models and dynamic agents, bridging text-visual distribution gaps while maintaining modality-specific features. This dual mechanism addresses conventional limitations and enables adaptive alignment through evolving feature distributions.

Taking text-to-video retrieval as an example, we obtain $\mathbf{v}_d$ and $\mathbf{t}_d$ using Stochastic Domain Modeling. All text stochastic domain embeddings $\mathbf{t}_d$ in a set of B samples are concatenated to form a tensor $\mathbf{T}_d \in \mathbb{R}^{B \times d}$, while video stochastic domain embeddings are similarly concatenated into a tensor $\mathbf{V}_d \in \mathbb{R}^{B \times d}$. The required query (Q), key (K), value (V), and agent (A) for agent attention are generated via noise-aware projection as shown in Figure 2:

$$\begin{aligned} Q_t &= \mathbf{W}_{Q_t}(\mathbf{T}_d + \text{Noise}(k)), \\ K_v &= \mathbf{W}_{K_v}(\mathbf{V}_d + \text{Noise}(k)), \\ V_v &= \mathbf{W}_{V_v}(\mathbf{V}_d + \text{Noise}(k)), \\ A_q &= \text{Pooling}(Q_t), \\ A_k &= \text{Pooling}(K_v), \end{aligned} \quad (9)$$

where $\text{Noise}(k)$ encodes noise level k into a d -dimensional embedding. $\mathbf{W}_{Q_t}, \mathbf{W}_{K_v}, \mathbf{W}_{V_v}$ are projection matrices, and A_q is dynamically generated through pooling query features. Pooling acts as a downsampling operation, projecting $Q_t \in \mathbb{R}^{B \times d}$ into a lower-dimensional space $\mathbb{R}^{b \times d}$. Similarly, A_k is generated by pooling key

features K_v . Additionally, the agent can be initialized as a static low-dimensional tensor, as detailed in Table 4.

To enhance semantic alignment between queries and candidates, we design a dual-branch attention mechanism. By utilizing the agent token, our approach enables both query-agent and agent-candidate alignment while improving computational efficiency. The complexity of agent attention [19] is $O(Bbd)$, where B represents the number of samples in the set, with $b \ll B$, significantly lower than standard attention $O(B^2d)$ [52]. The cross-modal data distribution $\mathcal{E}^A$ is computed as follows:

$$\mathcal{E}^A = \underbrace{\left(\text{Softmax}(Q_t A_k^T) \right)}_{\text{Query-Agent Alignment}} \cdot \underbrace{\left(\text{Softmax}(A_q K_v^T) \right)}_{\text{Agent-Candidate Alignment}} + X_k V_v + \text{DWC}(V_v) \quad (10)$$

Here, X_k is defined as a weighting factor that reflects the joint probability of previous noise levels for each video candidate, with higher values indicating greater confidence. By dynamically adjusting attention weights, our method enhances responses to high-confidence candidates. $\text{DWC}(\cdot)$ represents the depthwise convolution module [12], which preserves feature diversity.

We consider $\mathcal{E}^A$ as the high-level aligned distribution of text-video data and use it as input for the denoising decoder. The denoising decoder consists of a multilayer perceptron (MLP) [51] with a linear layer employing a ReLU [17] activation function for feature encoding, followed by another linear layer to compute the output distribution. Similarly, in video-to-text retrieval tasks, we treat the projection of the video representation as the query, and the projection of the text representation as the key and value in the attention module, computing the output distribution in the same manner.

During the reverse denoising process, expressed as $\hat{X}_t = \sqrt{\alpha_t} \hat{X}_{t-1} + \sqrt{1 - \alpha_t} \epsilon$, the predicted data distribution $\hat{x}_t$ is gradually generated. The loss function is designed to optimize noise prediction and data distribution modeling:

$$\mathcal{L}_G = \mathbb{E}_{(t,v) \in C} \left[\text{KL}(X_0 \| \phi(v, t, X_k)) \right] + \mathbb{E}_{(v,t) \in C} \left[\text{KL}(X_0 \| \phi(t, v, X_k)) \right] \quad (11)$$

Through the noise injection and denoising iteration process, the modeling of the joint distribution can fully capture the underlying semantic relationships between text-video pairs.

Furthermore, the generated stochastic embeddings t_d and v_d enhance the diffusion network’s capability to model the joint probability distribution. Simultaneously, by modeling the joint probability distribution, the diffusion model inversely optimizes the semantic alignment of text-video features, thereby forming a synergistic optimization loop. Notably, the proposed dual-branch attention mechanism introduces agent tokens to construct a parallel computational architecture comprising query-agent alignment and agent-candidate alignment: the former focuses on fine-grained cross-modal feature matching, while the latter models global relationships within the candidate set through agent nodes. This architecture not only significantly reduces the quadratic computational complexity of traditional cross-attention mechanisms [20, 52], but also enhances the model’s ability to capture long-tail semantic correlations through the information routing function of agent nodes [19, 21].

4 Experiments

This section begins by detailing the experimental setup, including the datasets, evaluation metrics, and implementation specifics. Subsequently, the experimental results are presented and compared with existing baseline methods. Next, the key model designs are analyzed, followed by ablation studies. Finally, the effectiveness of the proposed method is validated through visual analysis and case studies.

4.1 Experimental Settings

Datasets. This study is evaluated on five benchmark datasets: (1) MSRVT [59] consists of 10,000 YouTube videos, each with 20 manually annotated textual descriptions. Following the 1k-A split protocol [40], 9,000 videos are used for training and 1,000 videos for testing. (2) LSMDC [50] contains 118,081 video clips from 202 movies, with 1,000 test samples divided according to the standards of [16]. (3) DiDeMo [3] includes 10,642 video clips and 40,543 textual descriptions, using the data split standard of [28, 44]. (4) ActivityNet [31] consists of 20,000 YouTube videos, with the “val1” split [34, 55] comprising 10,009 training videos and 4,917 testing videos. We report results on the “val1” split of 10,009 and 4,917 as the train and test set. (5) VATEX [54] contains 34,991 video clips, each with multi-dimensional manually annotated textual descriptions.

Metrics. The retrieval performance is evaluated using the following metrics: Recall at K (R@K) for retrieval accuracy at ranks @1,5,10, Median Rank (Mdr), and Mean Rank (MnR).

Implementation Details. Based on the CLIP framework [49], we employ a dual-backbone architecture with ViT-B/32 and ViT-B/16 for cross-modal feature alignment. The input dimensions are standardized as follows: text sequence length of 32, video frame sequence of 12, and spatial resolution of 224×224 . We adopt a two-stage optimization strategy: Stage 1 uses a contrastive learning paradigm, initializing the stochastic domain module with a learning rate of $1e-5$, text/video encoders with $1e-6$, weight decay of 0.2, and Dropout of 0.3. The training is performed for 5 epochs (batch size 128) using the Adam optimizer [29], with a cosine annealing scheduler [42]. Stage 2 optimizes the denoising network based on a diffusion model, setting the text/video encoder learning rate to $1e-7$, other modules to $1e-3$, temperature parameter $\hat{\tau}=0.01$. The experiments are conducted on an NVIDIA A40 GPU platform. For both text-to-video and video-to-text retrieval tasks, we assume that only the candidate set is known beforehand. During the inference phase, we consider both the distance between video and text in the feature space and their joint probability distribution.

4.2 Performance Comparison

Baselines. To systematically validate the superiority of DomainDiff, we conduct comprehensive comparisons with state-of-the-art methods in text-video retrieval, categorized into four major groups: (1) mixed-granularity methods based on CLIP pre-trained models (X-pool [18], TS2-Net [41], UATVR [15], TMASS [53], EMCL-Net [25], UCoFiA [57], DiCoSA [27], HBI [26]); (2) modality fusion methods incorporating audio captions (VAST [8], VALOR [39], vid-TLDR [11]); (3) two-stage hybrid training frameworks (UMT [32], DiffusionRet [28]); and (4) unified multimodal task large models (UMT [32], VAST [8], VALOR [39], vid-TLDR [11]). Notably, we also

Methods	Text-to-Video					Video-to-Text				
	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓	R@1↑	R@5↑	R@10↑	MdR↓	MnR↓
EMCL-Net [25] NeurIPS22	46.8	73.1	83.1	2.0	12.8	46.5	73.5	83.5	2.0	8.8
X-Pool [18] CVPR22	46.9	72.8	82.2	2.0	14.3	44.4	73.3	84.0	2.0	9.0
TS2-Net [41] ECCV22	47.0	74.5	83.8	2.0	13.0	45.3	74.1	83.7	2.0	9.2
DiCoSA [27] IJCAI23	47.5	74.7	83.8	2.0	13.2	46.7	75.2	84.3	2.0	8.9
HBI [26] CVPR23	48.6	74.6	83.4	2.0	12.0	46.8	74.3	84.3	2.0	8.9
DiffusionRet [28] ICCV2023	49.0	75.2	82.7	2.0	12.1	47.7	73.8	84.5	2.0	8.8
UCoFiA [57] ICCV2023	49.4	72.1	83.5	–	–	–	–	–	–	–
UATVR [15] ICCV2023	50.8	76.3	85.5	1.0	12.4	48.1	76.3	85.4	2.0	8.0
UMT-L [32] ICCV2023	58.8	81.0	87.1	1.0	–	58.6	81.6	86.5	1.0	–
VAST [8] NeurIPS2023	63.9	84.3	89.6	1.0	–	–	–	–	–	–
T-MASS [53] CVPR2024	50.2	75.3	85.1	1.0	11.9	47.7	78.0	86.3	2.0	8.0
vid-TLDR [11] CVPR2024	58.1	81.0	86.8	1.0	–	58.7	81.6	86.9	1.0	–
VALOR [39] TPAMI2024	59.9	83.5	89.6	1.0	–	–	–	–	–	–
GRAM [13] ICLR2025	64.0	–	89.3	1.0	–	64.8	–	91.5	1.0	–
DomainDiff(ViT-32)	66.7	87.5	91.5	1.0	5.9	56.4	80.9	86.4	1.0	3.9
DomainDiff(ViT-16)	69.2	89.5	93.8	1.0	4.3	61.7	87.9	90.6	1.0	3.4

Table 1: Comparisons to current state-of-the-art methods on the MSRVTT dataset. “↑” denotes that higher is better. “↓” denotes that lower is better. Bold denotes the best performance. “–”: result is unavailable

Method	LSMDC Retrieval					DiDeMo Retrieval				
	R@1 ↑	R@5 ↑	R@10 ↑	MdR ↓	MnR ↓	R@1 ↑	R@5 ↑	R@10 ↑	MdR ↓	MnR ↓
UATVR [15] ICCV2023	–	–	–	–	–	45.8	73.7	83.3	2.0	13.5
DiffusionRet [28] ICCV2023	24.4	43.1	54.3	8.0	40.7	46.7	74.7	82.7	2.0	14.3
CLIP-ViP [60] ICLR2023	29.4	50.6	59.0	5.0	–	50.5	78.4	87.1	1.0	–
T-MASS [53] CVPR2024	30.3	52.2	61.3	5.0	40.1	53.3	80.1	87.7	1.0	9.8
vid-TLDR [11] CVPR2024	32.4	53.3	63.2	–	–	62.4	86.2	91.6	1.0	–
UMT-L [32] ICCV2023	37.7	60.6	67.3	–	–	59.7	84.9	90.8	1.0	–
VALOR [39] TPAMI2024	34.2	56.0	64.1	–	–	61.5	85.3	90.4	1.0	–
DomainDiff(ViT-32)	40.3	65.4	74.6	2.0	16.6	56.1	83.4	90.8	1.0	7.2
DomainDiff(ViT-16)	45.1	71.1	79.9	2.0	12.6	62.7	86.1	90.8	1.0	6.4

Method	ActivityNet Retrieval					VATEX Retrieval				
	R@1 ↑	R@5 ↑	R@10 ↑	MdR ↓	MnR ↓	R@1 ↑	R@5 ↑	R@10 ↑	MdR ↓	MnR ↓
vid-TLDR [11] CVPR2024	66.7	88.6	94.4	1.0	–	–	–	–	–	–
UMT [32] ICCV2023	66.8	89.1	94.9	1.0	–	72.0	95.1	97.8	1.0	–
VALOR [39] TPAMI2024	70.1	90.8	95.3	1.0	–	78.5	97.1	98.7	1.0	–
VAST [8] NeurIPS2023	70.5	90.9	95.5	1.0	–	83.0	98.2	99.2	1.0	–
GRAM [13] ICLR2025	69.9	–	96.1	1.0	–	87.7	–	100	1.0	–
DomainDiff(ViT-32)	70.3	90.2	95.3	1.0	2.8	83.2	98.0	98.9	1.0	1.8
DomainDiff(ViT-16)	72.3	91.0	95.8	1.0	1.9	85.7	98.6	99.4	1.0	1.5

Table 2: Text-to-Video retrieval performance compared to current SOTA methods on other datasets. “↑” denotes that higher is better. “↓” denotes that lower is better. Bold denotes the best performance. “–”: result is unavailable.

compare with the recently proposed GRAM [13] method, which achieves cross-modal alignment by optimizing the Gramian determinant and enhances performance by incorporating additional modal information.

Performance. As shown in Tables 1 and 2, DomainDiff demonstrates significant advantages across five benchmark datasets. For Text-to-Video (T2V) retrieval, experimental results indicate that on the MSRVTT [59] dataset, DomainDiff achieves a 5.2% performance improvement relative to the latest method, GRAM; and on

the LSMDC [50] dataset, it attains a 7.4% improvement compared to the previously best-performing UMT-L [32]. Notably, although GRAM [13] optimizes by incorporating additional modal features through the construction of an n-dimensional parallel polyhedron, and UMT-L employs a large-scale ViT-L/16 architecture, DomainDiff still achieves a breakthrough in performance under a more concise architecture. This performance improvement is primarily attributed to the synergistic effect between its innovative stochastic

Domain $\mathcal{D}$	Text-to-Video				
	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
$\exp(\frac{1}{M} \sum S_i)$	53.2	77.6	85.6	1.0	10.7
$\exp(\frac{\theta}{M} \sum S_i)$	53.9	77.8	85.6	1.0	10.5
$\exp(\mathbf{S}\Theta)$	56.8	80.7	86.2	1.0	9.8

Table 3: Model Discussion on Domain Modeling Design. We conduct experiments on MSRVT using a ViT-B/32 video encoder. The final performance comparison is based on choosing $\exp(\mathbf{S}\Theta)$.

Agent	LSMDC			DiDeMo		
	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$
static	44.2	69.9	79.5	61.2	85.7	90.2
dynamic	45.1	71.1	79.9	62.7	86.1	90.8

Table 4: Model Discussion on Agent Token Design. We conduct experiments on the LSMDC and DiDeMo datasets using a ViT-B/16 video encoder, with experimental results focusing on Text-to-Video tasks. The final performance comparison is based on the selection of the dynamic agent token.

domain embedding mechanism and the joint probability distribution modeling of the diffusion network. Moreover, we observed that, compared with the video encoder based on ViT-B/32, the one based on ViT-B/16 consistently exhibits improved performance across all datasets, indicating that finer-grained features are beneficial to the model.

However, as shown in Table 2, on the VATEX [54] dataset, the GRAM [13] method shows slightly higher performance. Considering that GRAM incorporates additional modal information, while our study avoids extra models to save computational cost, these results are not directly comparable. Furthermore, as shown in Table 1, there is a modest difference between the accuracies of Video-to-Text (V2T) and Text-to-Video(T2V) retrieval. It is inferred that for some short videos, the limited amount of information in both the video and text constrains semantic exploration during stochastic domain sampling, making it challenging to achieve significant performance improvements.

In summary, DomainDiff achieves progressive representation refinement via a two-stage joint optimization strategy: In the first stage, stochastic domain modeling provides a stable initial distribution for the diffusion network; in the second stage, the reverse denoising process dynamically optimizes the stochastic domain range($\mathcal{D}_t, \mathcal{D}_v$) via gradient backpropagation, thereby forming a closed-loop optimization mechanism. We will conduct a more in-depth analysis of this mechanism in subsequent sections.

4.3 Model Discussion

Stage I Calculation of Domain. As shown in Table 3, we proposed and compared three schemes: (1) $\exp(\frac{1}{M} \sum S_i)$: Uses fixed parameters and lacks learnability; (2) $\exp(\frac{\theta}{M} \sum S_i)$: Introduces a learnable scalar θ ; (3) $\exp(\mathbf{S}\Theta)$: Employs a linear transformation matrix $\mathbf{W}$. The experiments show that the $\exp(\mathbf{S}\Theta)$ scheme performs the best, benefiting from its stronger nonlinear modeling capability. Additionally, we found that using a single learnable scalar also achieves strong performance, indicating that the stochastic domain module is not highly sensitive to the design of $\mathcal{D}$. Ultimately, the $\exp(\mathbf{S}\Theta)$ scheme is selected.

(a) Effect of Training Strategy.

Training Strategy		Text-to-Video			
Stage1 $\mathcal{L}_{ce}$	Stage2 $\mathcal{L}_G$	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MnR $\downarrow$
$\checkmark$		56.8	80.7	86.2	9.8
	$\checkmark$	46.2	73.5	83.1	13.1
$\checkmark$	$\checkmark$	69.2	89.5	93.8	4.3

(b) Effect of Diffusion Steps on R@1 in Stage II.

Train	Eval			
	10	50	100	500
10	68.3	x	x	x
50	68.7	69.2	x	x
100	67.3	68.2	67.4	x
500	67.5	67.9	68.0	68.0

(c) Effect of Stochastic Sampling Frequency on Stage I.

Frequency (F)	R@1 $\uparrow$	R@5 $\uparrow$	R@10 $\uparrow$	MdR $\downarrow$	MnR $\downarrow$
5	50.4	75.6	85.4	1.0	12.1
10	55.4	78.4	85.8	1.0	11.1
15	56.8	80.7	86.2	1.0	9.8
20	56.2	80.2	86.0	1.0	10.0

Table 5: Ablation Experiments on the MSRVT Dataset for Text-to-Video tasks. “ $\mathcal{L}_G$ ” denotes Kullback-Leibler divergence, and “ $\mathcal{L}_{ce}$ ” denotes Cross-Entropy Loss. “ $\checkmark$ ” indicates that the stage of training is conducted, while “x**” indicates a failure in the evaluation process. Default settings are highlighted in blue .**

Stage II Initialization of Agent tokens. As shown in Table 4, we compared two strategies: (1) Static Strategy: Initializes agent tokens using fixed-dimensional zero tensors; (2) Dynamic Strategy: Generates low-dimensional tensors through dynamic downsampling based on Q, K . The experimental results demonstrate that the dynamic strategy exhibits superior performance and faster convergence on most datasets, as the agent tokens can directly capture semantic information from Q, K . Therefore, the dynamic initialization strategy is ultimately selected.

4.4 Ablation Studies

Training Strategy. We systematically compared various training strategies in Table 5a. The experimental results indicate that using a single domain-embedding-based contrastive learning strategy outperforms pure diffusion learning based on denoising networks, while the proposed two-stage hybrid training strategy achieves significant performance improvements. This validates that an excellent representation encoder can effectively unlock the potential of the diffusion denoising network, and demonstrates that the two-stage hybrid training strategy successfully combines the advantages of both approaches.

Analysis of Diffusion Steps. We analyzed the impact of the number of diffusion steps on model performance in Table 5b. The experimental results reveal that 50 diffusion steps achieve optimal performance in the text-video retrieval task, which is significantly lower

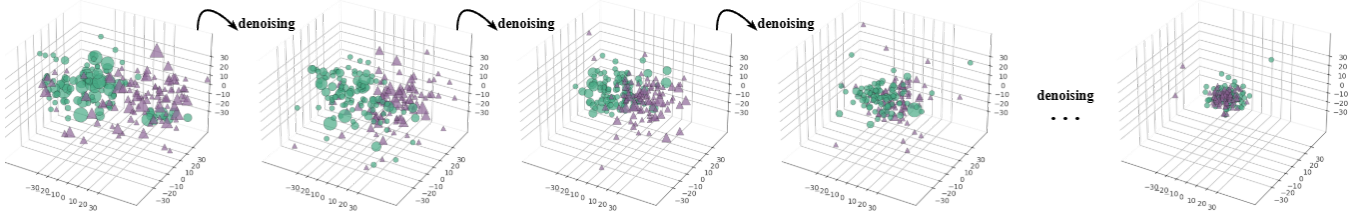


Figure 3: As denoising progresses, the text and video embeddings in the feature space gradually converge. **Triangles** represent text embeddings, and **circles** represent video embeddings. The size of the shapes corresponds to $\mathcal{D}_t, \mathcal{D}_v$, which decreases over time, indicating a more refined exploration of the semantic space.



Figure 4: Case study of our model in the text-video retrieval task. The orange box indicates that the video with the highest similarity retrieved by our model is the ground truth.

than the 1000 steps typically used in image generation tasks [14, 23]. We attribute this to the lower complexity of the distribution required for cross-modal alignment in text-video retrieval tasks compared to the complex pixel-level distribution in image generation tasks. Fewer diffusion steps are sufficient to capture the necessary semantic information.

Domain Sampling Frequency. As shown in Table 5c, we investigated the influence of the domain sampling frequency F during the inference phase. The experiments demonstrate that as F increases from 5 to 15, multiple stochastic samplings of t_d and v_d can explore better semantic representations, thereby improving performance. When $F = 5$, performance significantly drops, indicating that insufficient sampling frequency cannot fully explore the semantic space. Notably, performance stabilizes in the range of $F = 10$ to $F = 20$. Balancing performance and computational cost, the final model selects $F = 15$.

4.5 Qualitative Analysis

Figure 3 visualizes the dynamic evolution of cross-modal alignment during the reverse denoising process. As denoising iterations progress, the embedding distributions of text and video gradually converge, with the distances between corresponding pairs significantly decreasing. Meanwhile, the domain ranges ($\mathcal{D}_t$ and $\mathcal{D}_v$, indicated by the size of the shapes) continuously contract, reflecting increasingly precise semantic exploration and ultimately leading to consistent high-level semantic representations.

To further assess the retrieval performance of our model in real scenarios, we conduct a case study using representative samples from the test set. These cases typically feature videos with redundant frames and concise textual descriptions, posing a challenge for alignment. As illustrated in Figure 4, our model consistently

retrieves the ground-truth video as the top-1 result. Additionally, the top-k retrieved videos demonstrate strong semantic relevance to the queries, validating the effectiveness of our denoising-based alignment strategy.

5 Conclusion

This paper tackles the key challenge of cross-modal semantic alignment in text-video retrieval by introducing the DomainDiff framework. It aims to address the gap between the conciseness of textual semantics and the redundancy of video information, while addressing the performance limitations imposed by modality-specific representation differences in traditional fixed-point embedding approaches. Through a two-stage optimization process involving stochastic domain modeling and the dynamic agent attention diffusion network, DomainDiff not only strengthens the robustness of text and video semantic representations but also redefines cross-modal alignment by considering joint probability distributions. This provides a more flexible and efficient solution for text-video retrieval tasks. Experimental results demonstrate significant performance gains on several benchmark datasets, confirming the method’s superiority and generalization ability in complex scenarios.

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References

- [1] Hassan Akbari, Liangzhe Yuan, Rui Qian, Wei-Hong Chuang, Shih-Fu Chang, Yin Cui, and Boqing Gong. 2021. Vatt: Transformers for multimodal self-supervised learning from raw video, audio and text. *Advances in Neural Information Processing Systems* 34 (2021), 24206–24221.
- [2] Ahmed Alaa, Boris Van Breugel, Evgeny S Saveliev, and Mihaela Van Der Schaar. 2022. How faithful is your synthetic data? sample-level metrics for evaluating and auditing generative models. In *International Conference on Machine Learning*. PMLR, 290–306.
- [3] Lisa Anne Hendricks, Oliver Wang, Eli Shechtman, Josef Sivic, Trevor Darrell, and Bryan Russell. 2017. Localizing moments in video with natural language. In *IEEE International Conference on Computer Vision*. 5803–5812.
- [4] Max Bain, Arsha Nagrani, Gül Varol, and Andrew Zisserman. 2021. Frozen in time: A joint video and image encoder for end-to-end retrieval. In *IEEE/CVF International Conference on Computer Vision*. 1728–1738.
- [5] JM Bernardo, MJ Bayarri, JO Berger, AP Dawid, D Heckerman, AFM Smith, and M West. 2007. Generative or discriminative? getting the best of both worlds. *Bayesian statistics* 8, 3 (2007), 3–24.
- [6] Simion-Vlad Bogolin, Ioana Croitoru, Hailin Jin, Yang Liu, and Samuel Albanie. 2022. Cross modal retrieval with querybank normalisation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5194–5205.
- [7] Allen Chang, Matthew C Fontaine, Serena Booth, Maja J Matarić, and Stefanos Nikolaidis. 2024. Quality-diversity generative sampling for learning with synthetic data. In *AAAI Conference on Artificial Intelligence*, Vol. 38. 19805–19812.
- [8] Sihan Chen, Handong Li, Qunbo Wang, Zijia Zhao, Mingzhen Sun, Xinxin Zhu, and Jing Liu. 2024. Vast: A vision-audio-subtitle-text omni-modality foundation model and dataset. *Advances in Neural Information Processing Systems* 36 (2024).
- [9] Shizhe Chen, Yida Zhao, Qin Jin, and Qi Wu. 2020. Fine-grained video-text retrieval with hierarchical graph reasoning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 10638–10647.
- [10] Xing Cheng, Hezheng Lin, Xiangyu Wu, Fan Yang, and Dong Shen. 2021. Improving Video-Text Retrieval by Multi-Stream Corpus Alignment and Dual Softmax Loss. *CoRR* abs/2109.04290 (2021). arXiv:2109.04290
- [11] Joonmyung Choi, Sanghyeok Lee, Jaewon Chu, Minhyuk Choi, and Hyunwoo J Kim. 2024. vid-TLDR: Training Free Token merging for Light-weight Video Transformer. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 18771–18781.
- [12] François Chollet. 2017. Xception: Deep learning with depthwise separable convolutions. In *IEEE Conference on Computer Vision and Pattern Recognition*. 1251–1258.
- [13] Giordano Cicchetti, Eleonora Grassucci, Luigi Sigillo, and Danilo Comminiello. 2024. Gramian Multimodal Representation Learning and Alignment. *arXiv preprint arXiv:2412.11959* (2024).
- [14] Prafulla Dhariwal and Alexander Nichol. 2021. Diffusion models beat gans on image synthesis. *Advances in Neural Information Processing Systems* 34 (2021), 8780–8794.
- [15] Bo Fang, Wenhao Wu, Chang Liu, Yu Zhou, Yuxin Song, Weiping Wang, Xiangbo Shu, Xiangyang Ji, and Jingdong Wang. 2023. Uatvr: Uncertainty-adaptive text-video retrieval. In *IEEE/CVF International Conference on Computer Vision*. 13723–13733.
- [16] Valentin Gabeur, Chen Sun, Karteek Alahari, and Cordelia Schmid. 2020. Multi-modal transformer for video retrieval. In *European Conference on Computer Vision*. Springer, 214–229.
- [17] Xavier Glorot, Antoine Bordes, and Yoshua Bengio. 2011. Deep sparse rectifier neural networks. In *the 14th international conference on artificial intelligence and statistics*. JMLR Workshop and Conference Proceedings, 315–323.
- [18] Satya Krishna Gorti, Noël Vouitsis, Junwei Ma, Keyvan Golestan, Maksims Volkovs, Animesh Garg, and Guangwei Yu. 2022. X-Pool: Cross-Modal Language-Video Attention for Text-Video Retrieval. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 5006–5015.
- [19] Dongchen Han, Tianzhu Ye, Yizeng Han, Zhuofan Xia, Siyuan Pan, Pengfei Wan, Shiji Song, and Gao Huang. 2024. Agent attention: On the integration of softmax and linear attention. In *European Conference on Computer Vision*. Springer, 124–140.
- [20] Yizeng Han, Gao Huang, Shiji Song, Le Yang, Honghui Wang, and Yulin Wang. 2021. Dynamic neural networks: A survey. *IEEE transactions on pattern analysis and machine intelligence* 44, 11 (2021), 7436–7456.
- [21] Yizeng Han, Zeyu Liu, Zhihang Yuan, Yifan Pu, Chaofei Wang, Shiji Song, and Gao Huang. 2024. Latency-aware unified dynamic networks for efficient image recognition. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).
- [22] Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollár, and Ross Girshick. 2022. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16000–16009.
- [23] Jonathan Ho, Ajay Jain, and Pieter Abbeel. 2020. Denoising Diffusion Probabilistic Models. In *Annual Conference on Neural Information Processing Systems(2020)*.
- [24] Sarah Ibrahim, Xiaohang Sun, Pichao Wang, Amanmeet Garg, Ashutosh Sanan, and Mohamed Omar. 2023. Audio-enhanced text-to-video retrieval using text-conditioned feature alignment. In *IEEE/CVF International Conference on Computer Vision*. 12054–12064.
- [25] Peng Jin, Jinfa Huang, Fenglin Liu, Xian Wu, Shen Ge, Guoli Song, David Clifton, and Jie Chen. 2022. Expectation-maximization contrastive learning for compact video-and-language representations. *Advances in Neural Information Processing Systems* 35 (2022), 30291–30306.
- [26] Peng Jin, Jinfa Huang, Pengfei Xiong, Shangxuan Tian, Chang Liu, Xiangyang Ji, Li Yuan, and Jie Chen. 2023. Video-text as game players: Hierarchical banzhaf interaction for cross-modal representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2472–2482.
- [27] Peng Jin, Hao Li, Zesen Cheng, Jinfa Huang, Zhennan Wang, Li Yuan, Chang Liu, and Jie Chen. 2023. Text-Video Retrieval with Disentangled Conceptualization and Set-to-Set Alignment. In *International Joint Conference on Artificial Intelligence*. ijcai.org, 938–946.
- [28] Peng Jin, Hao Li, Zesen Cheng, Kehan Li, Xiangyang Ji, Chang Liu, Li Yuan, and Jie Chen. 2023. DiffusionRet: Generative Text-Video Retrieval with Diffusion Model. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2470–2481.
- [29] Diederik P. Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*.
- [30] Diederik P. Kingma and Max Welling. 2014. Auto-Encoding Variational Bayes. In *International Conference on Learning Representations*.
- [31] Ranjay Krishna, Kenji Hata, Frederic Ren, Li Fei-Fei, and Juan Carlos Niebles. 2017. Dense-captioning events in videos. In *IEEE International Conference on Computer Vision*. 706–715.
- [32] Kunchang Li, Yali Wang, Yizhuo Li, Yi Wang, Yanan He, Limin Wang, and Yu Qiao. 2023. Unmasked teacher: Towards training-efficient video foundation models. In *IEEE/CVF International Conference on Computer Vision*. 19948–19960.
- [33] Pandeng Li, Chen-Wei Xie, Hongtao Xie, Liming Zhao, Lei Zhang, Yun Zheng, Deli Zhao, and Yongdong Zhang. 2023. MomentDiff: Generative Video Moment Retrieval from Random to Real. In *Advances in Neural Information Processing Systems*, Vol. 36. 65948–65966.
- [34] Pandeng Li, Hongtao Xie, Jiannan Ge, Lei Zhang, Shaobo Min, and Yongdong Zhang. 2022. Dual-stream knowledge-preserving hashing for unsupervised video retrieval. In *European Conference on Computer Vision*. 181–197.
- [35] Yazhe Li, Jorg Bornschein, and Ting Chen. 2024. Denoising Autoregressive Representation Learning. In *International Conference on Machine Learning*.
- [36] Chen Liang, Wenguan Wang, Jiaxu Miao, and Yi Yang. 2022. Gmmseg: Gaussian mixture based generative semantic segmentation models. *Advances in Neural Information Processing Systems* 35 (2022), 31360–31375.
- [37] Chengzhi Lin, Ancong Wu, Junwei Liang, Jun Zhang, Wenhang Ge, Wei-Shi Zheng, and Chunhua Shen. 2022. Text-adaptive multiple visual prototype matching for video-text retrieval. *Advances in Neural Information Processing Systems* 35 (2022), 38655–38666.
- [38] Yan-Bo Lin, Jie Lei, Mohit Bansal, and Gedas Bertasius. 2022. Eclipse: Efficient long-range video retrieval using sight and sound. In *European Conference on Computer Vision*. Springer, 413–430.
- [39] Jing Liu, Sihan Chen, Xingjian He, Longteng Guo, Xinxin Zhu, Weining Wang, and Jinhui Tang. 2025. VALOR: Vision-Audio-Language Omni-Perception Pretraining Model and Dataset. *IEEE Trans. Pattern Anal. Mach. Intell.* 47, 2 (2025), 708–724.
- [40] Yang Liu, Samuel Albanie, Arsha Nagrani, and Andrew Zisserman. 2019. Use What You Have: Video Retrieval Using Representations From Collaborative Experts. *CoRR* abs/1907.13487 (2019). arXiv:1907.13487
- [41] Yuqi Liu, Pengfei Xiong, Luhui Xu, Shengming Cao, and Qin Jin. 2022. Ts2-net: Token shift and selection transformer for text-video retrieval. In *European conference on computer vision*. 319–335.
- [42] Ilya Loshchilov and Frank Hutter. 2017. Sgdr: Stochastic Gradient Descent with Warm Restarts. In *International Conference on Learning Representations*.
- [43] Dezhao Luo, Shaogang Gong, Jiabo Huang, Hailin Jin, and Yang Liu. 2024. Generative Video Diffusion for Unseen Cross-Domain Video Moment Retrieval. *CoRR* abs/2401.13329 (2024). arXiv:2401.13329
- [44] Huaishao Luo, Lei Ji, Ming Zhong, Yang Chen, Wen Lei, Nan Duan, and Tianrui Li. 2022. CLIP4Clip: An empirical study of CLIP for end to end video clip retrieval and captioning. *Neurocomputing* 508 (2022), 293–304.
- [45] Yiwei Ma, Guohai Xu, Xiaoshuai Sun, Ming Yan, Ji Zhang, and Rongrong Ji. 2022. X-clip: End-to-end multi-grained contrastive learning for video-text retrieval. In *the 30th ACM international conference on multimedia*. 638–647.
- [46] Antoine Miech, Ivan Laptev, and Josef Sivic. 2018. Learning a Text-Video Embedding from Incomplete and Heterogeneous Data. *CoRR* abs/1804.02516 (2018). arXiv:1804.02516
- [47] Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjeon Choi, and Jaewon Yoo. 2020. Reliable fidelity and diversity metrics for generative models. In *International conference on machine learning*. 7176–7185.
- [48] Jesús Andrés Portillo-Quintero, José Carlos Ortiz-Bayliss, and Hugo Terashima-Marin. 2021. A straightforward framework for video retrieval using clip. In *Mexican Conference on Pattern Recognition*. Springer, 3–12.
- [49] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning Transferable Visual Models

- From Natural Language Supervision. In *International Conference on Machine Learning*, Vol. 139. 8748–8763.
- [50] Anna Rohrbach, Marcus Rohrbach, Niket Tandon, and Bernt Schiele. 2015. A dataset for movie description. In *IEEE Conference on Computer Vision and Pattern Recognition*. 3202–3212.
 - [51] Ilya O Tolstikhin, Neil Houlsby, Alexander Kolesnikov, Lucas Beyer, Xiaohua Zhai, Thomas Unterthiner, Jessica Yung, Andreas Steiner, Daniel Keysers, Jakob Uszkoreit, et al. 2021. Mlp-mixer: An all-mlp architecture for vision. *Advances in Neural Information Processing Systems* 34 (2021), 24261–24272.
 - [52] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. *Advances in Neural Information Processing Systems* 30 (2017).
 - [53] Jiamian Wang, Guohao Sun, Pichao Wang, Dongfang Liu, Sohail Dianat, Majid Rabbani, Raghuveer Rao, and Zhiqiang Tao. 2024. Text Is MASS: Modeling as Stochastic Embedding for Text-Video Retrieval. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 16551–16560.
 - [54] Xin Wang, Jiawei Wu, Junkun Chen, Lei Li, Yuan-Fang Wang, and William Yang Wang. 2019. VaTeX: A Large-Scale, High-Quality Multilingual Dataset for Video-and-Language Research. In *IEEE/CVF International Conference on Computer Vision (ICCV)*.
 - [55] Xiaohan Wang, Linchao Zhu, and Yi Yang. 2021. T2vlad: global-local sequence alignment for text-video retrieval. In *IEEE/CVF conference on Computer Vision and Pattern Recognition*. 5079–5088.
 - [56] Ziyue Wang, Aozhu Chen, Fan Hu, and Xirong Li. 2022. Learn to understand negation in video retrieval. In *the 30th ACM International Conference on Multimedia*. 434–443.
 - [57] Ziyang Wang, Yi-Lin Sung, Feng Cheng, Gedas Bertasius, and Mohit Bansal. 2023. Unified Coarse-to-Fine Alignment for Video-Text Retrieval. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2816–2827.
 - [58] Chen Wei, Karttikeya Mangalam, Po-Yao Huang, Yanghao Li, Haoqi Fan, Hu Xu, Huiyu Wang, Cihang Xie, Alan Yuille, and Christoph Feichtenhofer. 2023. Diffusion models as masked autoencoders. In *IEEE/CVF International Conference on Computer Vision*. 16284–16294.
 - [59] Jun Xu, Tao Mei, Ting Yao, and Yong Rui. 2016. Msr-vtt: A large video description dataset for bridging video and language. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 5288–5296.
 - [60] Hongwei Xue, Yuchong Sun, Bei Liu, Jianlong Fu, Ruihua Song, Houqiang Li, and Jiebo Luo. 2022. CLIP-ViP: Adapting Pre-trained Image-Text Model to Video-Language Representation Alignment. *CoRR* abs/2209.06430 (2022). arXiv:2209.06430
 - [61] Henghao Zhao, Kevin Qinghong Lin, Rui Yan, and Zechao Li. 2024. DiffusionVMR: Diffusion Model for Joint Video Moment Retrieval and Highlight Detection. arXiv:2308.15109
 - [62] Shuai Zhao, Linchao Zhu, Xiaohan Wang, and Yi Yang. 2022. Centerclip: Token clustering for efficient text-video retrieval. In *the 45th International ACM SIGIR Conference on Research and Development in Information Retrieval*. 970–981.
 - [63] Linchao Zhu and Yi Yang. 2020. Actbert: Learning global-local video-text representations. In *IEEE/CVF conference on Computer Vision and Pattern Recognition*. 8746–8755.